Embodied Sketching

Elena Márquez Segura*, Laia Turmo Vidal*, Asreen Rostami, Annika Waern***

*Uppsala University
Uppsala, Sweden
{elena.marquez; laia.turmo;
annika.waern}@im.uu.se

**Mobile Life Centre,
Stockholm University, Sweden
asreen@dsv.su.se

ABSTRACT

Designing bodily experiences is challenging. In this paper, we propose *embodied sketching* as a way of practicing design that involves understanding and designing for bodily experiences early in the design process. Embodied sketching encompasses ideation methods that are grounded in, and inspired by, the lived experience and includes the social and spatial settings as design resources in the sketching. Embodied sketching is also based on harnessing play and playfulness as the principal way to elicit creative physical engagement. We present three different ways to implement and use embodied sketching in the application domain of co-located social play. These include *bodystorming* of ideas, *co-designing with users*, and *sensitizing designers*. The latter helps to uncover and articulate significant, as well as novel embodied experiences, whilst the first two are useful for developing a better understanding of possible design resources.

Author Keywords

Ideation; Bodystorming; Embodied Sketching; Embodied Interaction; Design Methods; Sensitizing; Somaesthetics.

ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

INTRODUCTION

More than fifteen years ago, Djajadiningrat, Gaver, and Frens discussed the difficulty of designing when considering the aesthetic experience and interaction together [13]. As time has passed there has been plenty of research into how to fill this gap by establishing methods, guiding principles, and supporting techniques [1,14,20,28,58]. Recently, the call for methods that can tap into the aesthetic experience has been renewed with the development of technology that can recognize bodily gestures, movements, and physiological reactions, [30,38,41,42,70]. Nonetheless, as in many other domains, technology has advanced faster than research [16], and we have found ourselves surrounded by a plethora of research prototypes and commercial examples that use gestures and movement as input, using diverse controllers and input mechanisms. These have focused on recognizing and harnessing our everyday gestures [55], so that we can operate in a virtual world in a way that bears resemblance with how we act in our everyday life, in a "natural way". This presented a powerful narrative, from which followed a variety of movement-based interfaces [55]. In the application domain of interactive games in particular these interfaces were embraced by industry, researchers, and the gaming public [36,69]. However, this phenomenon soon lost momentum, as reflected in a significant decrease in sales [15,50].

The approach of instrumentalizing the body [30], and the technical limitations of the technology to capture and map body movements [2,40], have been highlighted as issues due to the constraints they imposed on the moving body [14, 20]. While we have long passed the period of technical limitations, we have not yet witnessed the foreseen paradigm shift. Body-controlled game experiences do not come off as exciting and engaging to players in the long run, and gesture-based interaction fails to offer an interesting experience once the novelty effect passes [73]. Sony acknowledges with this comment about the PlayStation Move controller: "[…] great tech, probably not so great applications so far […]"[71].

A positivistic understanding of action that has dominated the field underlies many of these issues [55]; movements have been seen as something that can be specified, represented, and modeled without considering the context where they take place. This makes the design problem an "engineering" one, and gestures have been considered as "an afterthought", once the technology in "in place" [73].

In this paper, we present a design practice that we call *embodied sketching*, which is intended to: i) support *ideation* rather than evaluation, ii) support the *inclusion of bodily experiences* early in the design process, before the construction of fully-functional prototypes, and iii) *spur creativity* by harnessing play and playfulness, in a way that is grounded in and inspired by the lived experience. By

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.
CHI'16, May 07 - 12, 2016, San Jose, CA, USA
Copyright is held by the owner/author(s). Publication rights licensed to ACM.
ACM 978-1-4503-3362-7/16/05...$15.00
DOI: http://dx.doi.org/10.1145/2858036.2858486

using the word *embodied* we foreground the embodied perspective from which we draw [16]. By using the word *sketching*, we wish to highlight that the methods are directed towards ideation as well as the exploratory, suggestive, and tentative character of the resulting sketches [6].

We illustrate the concept of embodied sketching through three case examples, which represent three design situations where embodied sketching can be useful: in *bodystorming* with designers, in *co-creating with users*, and in *sensitizing designers*. While the former two are useful for ideation related to the available design resources (including, but not restricted to, technology), the latter helps in uncovering and articulating qualities of a novel embodied experience that share traits with that for which designers design.

With this paper, we contribute to HCI by adding to the body of work that aims to translate the abstract theory of embodied interaction into design practice, in this case in the domain of design for movement and play.

BACKGROUND

Spurring creativity

There exist a plethora of methods for spurring creativity and ideation that are used with enthusiasm by practitioners [27] in Interaction Design (IxD) processes. This includes well-known practices such as classical brainstorming [56], personas, affinity maps, and various sketching methods. However, these are not always rigorously studied or validated [18]. Recent research efforts in IxD have been devoted to develop, study, and validate such methods [18].

Among these methods we find bodystorming [27,57,66], which has been characterized as a situated ideation method in which designers prototype in context, using enactment [66]. Schleicher et al. [66] named *user-case theater* a variation of bodystorming that involves actors, and props, to simulate the activity for which to design. Related but with a more evaluative focus are methods of *strong prototyping,* which replicate the end environment so that design prototypes can be tested [66]. Particularly interesting for this article, is the concept of *embodied storming*, a pre-brainstorming activity in which participants engage in simulations of an envisioned scenario, which they collaborate to sustain by enactment. Through enactment, the participants can gain personal insight into the situation for which they are to design.

Understanding the embodied experience

While all of the above are inspirational, they are not specifically purposed for the design of bodily activities.

Design scholars have emphasized that bodily experience and awareness are key when designing technologically supported aesthetic experiences [29,39,41,48,63,65,81]. Much of this work is theoretically grounded in somaesthetics [38], an interdisciplinary field that foregrounds the role of the body not only as an object of aesthetic representation, but as subjectivity for aesthetic appreciation [67]. The field embraces "bodily perception, performance and presentation" theories and practices [68]. Analytic somaesthetics in particular has provided a theoretical foundation to researchers and designers dealing with the lived, and felt experience. Its theories about "the nature of our bodily perceptions and practices and their function in our knowledge and construction of the world" [68] have been useful for explaining the importance of bodily perception and awareness. For example, in [29] Höök presents a detailed analysis of the experience of horseback riding, which yields interesting themes that are then translated into design considerations.

Other work in this area has relied on the pragmatic branch of somaesthetics, which concerns the postulation of disciplines and techniques for improving our somatic sensibilities, typically accompanied by a particular way of understanding and shaping of the body (e.g. Feldenkrais, Tai Chi). Recognized work in this domain includes Klooster's explorations on the usage of artefacts and the creation of cycles of attention to (and through) the body, and the artefact [32,81], and Schiphorst's suggestion of introducing a "connoisseur", in which the design process is supported by a facilitator with heightened somatic and empathic skills, who is able to resonate with the experience of the participants, connect, and guide them to access subtle qualities of bodily knowledge that would otherwise not be accessible to them [65]. Schiphorst works with a shift from a third-person "everyday-type" perspective on movement towards a first-person, inwards-type of awareness [63–65]. that runs through the work of Shusterman. Also Lee et al. emphasise somatic introspection methods as a way to heighten attention and interest [38].

HCI has to a lesser extent explored the use of practical somaesthetics, which is the third branch of somaesthetics. This is perhaps the most straightforward of them all, as it involves the actual "doing", the practice of a certain discipline such as what practitioners do when they go to a Feldenkrais class. In this article, we take the opportunity to include practical somaesthetics in one of our embodied sketching techniques.

Drawing from other fields

A crucial issue for enactment-based ideation processes is how to facilitate enactment. Performance anxiety and body perception issues may affect the participants' ability to engage in physical performance in a co-located social setting [9]. Methods from other creative fields have been used to mitigate these issues. The use of scenario-based and character creation techniques [4,54] for understanding of prospective scenarios [3] and users is well understood, as it is the use of performative techniques in design [33]. Such methods have been used also in participatory design [43] to allow designers to explore, communicate and test the prototypes in a fictional space. Improvisational methods adopted from theatre can bridge the gap between users and

designers [53], and have shown potential as a way to investigate the human behaviour in space, with or around objects [5]. Situated performance activities, including role playing and embodied fiction have been used as ways to co-design and generate creative ideas [34]. Experience Design Theatre (EDT) as introduced by Vines et al [77] is a method that investigates participants interaction in a shared lived experience through the use of role playing. Finally Gerber [24,25], uses improvisation techniques to support designers and improve brainstorming, by building up skills and a mindset that is productive during ideation phases. These techniques will often work to reinforce the imperatives of brainstorming (withholding judgment, building on one another's ideas, focus on quantity, etcetera), and contribute to foster innovation, spontaneity, and the ability to embrace failure. In this article, we rely on play activities in a similar way.

OUR APPROACH

In this section, we discuss the design values that underlie embodied sketching.

Design resources for co-located physical & social play

In the introduction we elaborated on how the design space of movement-based interfaces remains narrow. This is in part due to a lack of acknowledgement of how such interfaces shape the space around the technology, including the social and emotional experience of the players [35], but also in how shaping the space around an activity can influence the play experience. We see a missed opportunity in not considering that space as a design resource [46,47,73], in particular, the socio-spatial arrangement of players and artefacts in the space [46,47].

The technology has not yet caught up with this [69], and we find plenty of "playing, seating, and viewing arrangements that hinder mechanisms such as mutual eye contact, natural reciprocation of approach or avoidance cues and mirroring, or emotionally relevant communication signals" [47]. The "social factor", which is one of the main motivations why players like playing multiplayer games [9,17,22,78], remains often set aside as a backdrop, without having direct involvement, participation, and interaction in the gameplay. This has been known by the moniker of "playing alone together" [17,72,76] .

The potential of the socio-spatial configurations for shaping the situated experience is well documented [22,37,44,45]. Hence, although the mere presence of others can add to the game experience even in cases of solo play [7], play schemes along the lines of "alone together" miss the benefits associated with taking the social aspect into account when designing for co-located social play – among others, higher engagement, arousal, affect, and positive emotions [9,22,61].

In line with the concepts reported in [47], we take a technology-supported approach [79] and consider the technology to be just one of the resources to design. The others include socio-spatial elements, such as arrangements of players, or the placement of displays used for monitoring actions and effects. Our approach typically involves designing an activity by selecting an amalgam of design resources (e.g. digital and physical artefacts, participants, audience), each to whom/which an important role to support the activity is assigned. Their spatial arrangement is also designed, so that the whole configuration promotes mutual orientation, awareness, and communication and coordination of joint action [32,47].

A practice to access the experience

As discussed in the background section, techniques from other disciplines like play and theatre have been imported to spark a mindset that is conducive of creativity

We propose that practical somaesthetics can function as a way to access to, and articulate bodily experiences in a manner that informs design. . We suggest that designers prepare for ideation by engaging in an activity very similar to that for which they design, containing similar elements, and physical core mechanics[1]. This allows designers to develop a heightened embodied sensibility attentive to both the designers' somatic experience (intero-, proprio-, and exteroceptive sensations), as well as important contextual elements (e.g. social, physical, and spatial). We argue that this in turn inspires and influences the emergence of design ideas that take the embodied experience at their core.

Play as a medium and a goal

Play and playfulness have proven themselves useful in instilling a mindset that is conducive of creative thinking [12,27]. Play can also help to lower performance anxiety due to its power to re-signify [11,26,31,59] anything that happens within the "magic circle of play"[31]. As Gaver puts it: *play is not just mindless entertainment, but an essential way of engaging with and learning about our world and ourselves—for adults as well as children. As we toy with things and ideas, as we chat and daydream, we find new perspectives and new ways to create, new ambitions, relationships, and ideals.*[23].

Before, we mentioned how methods from improvisational theatre could positively impact ideation activities. We argue that play and playfulness can spark off creativity in very similar ways [23].

EMBODIED SKETCHING

We define *Embodied sketching* as a characterization of design practices in the domain of embodied interaction that

[1] In game studies, core mechanics are recurring or continuous actions and procedures required to advance in a game [62:316]. As we are designing for physical play, we introduce 'physical', to highlight those core mechanics that rely on substantial physical engagement by players. They can be simple actions (e.g. jumping when you are playing on a jump rope) or combination of these (e.g. shooting while hiding and dodging when playing laser tag).

foregrounds the somaesthetic experience for the exploration of, and design for particularly interesting physical activities. This is done by: 1) physically engaging with simple actions and activities that are constituent of, or essential to, the bodily activity you design for, and 2) by looking into (and trying) how different design resources integrate and blend together to support and give rise to play activities.

Embodied sketching can be implemented in different ways, depending on the current status of a design project and the particular challenges that it faces. Below, we present examples of embodied sketching in design projects, and show some of the considerations that come into play. We introduce our example cases with a short description of the design situation at hand, then describe implementation details, and finish with our main takeaways.

The first usage scenario, *bodystorming*, is used to help designers sketch ideas for movement-based interactive systems by engaging physically in a co-design play-based ideation activity with peers. The exploration of the design space focuses on both digital and non-digital mediating artefacts, on the arrangement of contextual elements in physical space, and on the 'physical core mechanics' at work. This form of embodied sketching allows designers to develop ideas from scratch.

The second scenario, *participatory embodied sketching*, uses play as a mean to support a co-design activity with end users. As a participatory form of embodied sketching, it democratizes design and empowers end users [60], supporting them to explore, follow, play, modify, break, and create game or activity design structures. This form of embodied sketching allows designers to understand their users, and to refine and rethink existing ideas.

Finally, the practice-driven *sensitizing of designers,* employs an activity to cultivate the somaesthetic appreciation of designers, to sensitize and help them access and articulate particularly interesting embodied phenomena we want to design for, and hence it can be employed pre-ideation.

Example 1: Bodystorming

In a joint international research collaboration project, we worked with a research group that had developed an exertion game – an interactive game that "deliberately requires intense physical effort" from the participants [51]. The central physical activity had already been decided – centred around the experience of hanging. An interesting design, "hanging off a bar" [52] had been created, challenging previous exertion games designs by facilitating "exertion" without requiring big and energetic movements.

At the time of the embodied sketching exercise, the design team had already collected feedback from different play tests. However, we thought that the very interesting concept and core mechanics of hanging had not been fully explored; we felt that hanging could offer a *generative concept* from which new game ideas could arise. Hence, we decided to re-open the design space focusing on the concept of hanging. We arranged a bodystorming workshop using a non-technological tool: the TRX, an artefact for suspension training that leverages body weight [75].

Method

An expert design team of the Exertion Games Lab at RMIT, composed of four Ph.D. students and one master's student in Game Design, participated in the workshop. The TRX was hung from the ceiling in the middle of the workshop space. We also brought other artefacts gathered from throughout the lab, such as Styrofoam swords, a Pilates ball, a basketball ball, tennis balls, and Styrofoam mats.

We introduced the concept of embodied sketching to the designers: we were to come up with games by playing with the material we had around. We enforced social rules that were similar to those of ordinary brainstorming [56] in the sense that all ideas were welcome, that we would not criticize ideas, and that we wouldn't think too much about feasibility issues. Then, we suggested turn-taking as to regulate social interaction: First, a person would suggest something, which s/he would show/demo to the rest of us. Then, the rest of the group was free to participate in the idea construction by suggesting changes, e.g. adding rules, goals, and artefacts. When one idea was "done", we would come up with another idea together. The workshop was recorded in order to later discuss the session with a senior designer and two of the participants.

Overview of games and ideas

Ten ideas arose from the workshop, which we named 1) ninja swords, 2) projections on the floor, 3) sitting on a ball, 4) standing on a ball, 5) flying football, 6) dodge balls, 7) laser, 8) competitive path following, 9) hitting targets and 10) single legged. Six of them included variations, from a minimum of three to a maximum of seven (1, 2, 4, 8, 9, and 10). Variations were adds-on to an original idea, with different physical core mechanics or artefacts involved.

Physical core mechanics and artefacts shaping ideas

The ideas that emerged throughout the workshop were primarily shaped by either physical core mechanics, or

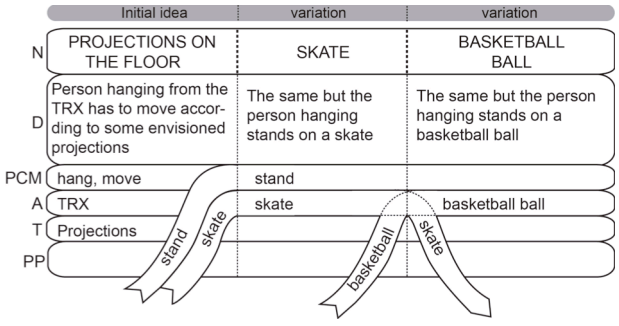


Fig. 1 © Laia Turmo Vidal & Elena Márquez Segura. Graphical representation of the evolution of one idea. N=name, D=description, PCM=physical core mechanics, A=artefacts, T=envisioned technology, PP=pretence play.

artefacts (see in Fig. 1. how different elements were added and removed, resulting into different variations of one same idea). Among the physical core mechanics that appeared, there were some that "worked", i.e. where enjoyment was visible, players showed great excitement, or an obvious wish to engage. An example of this was the idea "standing on a ball" (4), where one participant was hanging from the TRX while trying to maintain the balance on top of a Pilates ball. We engaged with the idea longer in comparison to the other ones, most of us tried it out, and the sound of laughter was present continuously.

At other times, the physical core mechanics needed some tweaking. This happened when the group identified some interesting elements in the idea but the idea itself did not "work". When this happened, the participants tended to replace an artefact to another that changed the physical core mechanic. This happened with one of the versions of the idea "projections on the floor" (2, see Fig. 1), where one participant was hanging from the TRX while standing on a skateboard, and envisioned certain projections on the floor which he should try to avoid. The skateboard turned out to constrain his movement too much. One of the participants stated: "*as long as the skateboard is there, we don't have that much of room*". In response to this it was suggested to change the skateboard for a basketball. This proved to be useful when another participant tried it out as it allowed a broader range of movements. Figure 1 depicts part of the evolution of the initial idea according to these changes[2].

Some physical core mechanics were discarded completely, i.e. when an idea emerged but was not accepted or soon abandoned. For instance, a participant suggested trying the idea "flying football" (5), where a participant would use the TRX to gather momentum, to then take her feet off the ground and hit a Pilates ball. After the first try by the proponent of the idea, the rest of the group showed no intention to try it out, change it, or build upon it. Thus it was discarded in favour of the next idea.

As with the physical core mechanics, the available artefacts also shaped the ideas. Some artefacts seemed to "work better" (in the same sense than used above) than others in particular ideas. For instance, during the exploration of the idea "standing on a ball" (4), one iteration was suggested involving tennis balls, which would be thrown to a player on the Pilates ball, who needed to dodge them. While the big ball was kept for the following ideas, the tennis balls were discarded after just one try. This is also an illustrative example of how an idea centred on an artefact (Pilates ball) allowed changes of the physical core mechanics involved (from just balance, to balance and dodging).

[2] The full evolution of all the ideas in this session can be found as auxiliary material of this paper at ACM DL.

Technology in the workshop

While this workshop was not centred on proposing technological implementations, this would still emerge as part of the game explored. One example was the idea called "projections with skateboard" (variation of 2), where the group envisioned some technology that would implement a particular feature of the game, like projecting patterns on the floor, which the participant on the skateboard holding the TRX had to follow or avoid.

In other ideas, the designers would simulate the functioning of the technology using *pretence play* [19] (PP in Fig.1). The role that the technology would take was simulated using the artefacts at hand, which promoted discussions about possible future issues. For example, in the idea called "hitting targets" (9), the group imagined the random appearance of dots in the environment, which the participants hanging from the TRX had to catch. The appearance of the dots was simulated by placing, and moving Styrofoam panels around. Playing this out allowed us to realize the need for creating some rules concerning the order of appearance of the dots, and inspired variations of the idea, such as moving the dots in the vertical plane, which was simulated by holding the panels up in the air.

Debrief

The debrief session revolved around how the TRX was a facilitator of bodily play, and the simple physical core mechanics that the TRX invited, such as hanging and swinging, from which other more complex arose. This led to a discussion about the different physical affordances of the TRX as an artefact, and the bodily impact of using it in different ways. E.g. the muscles used depending on the grip, position, and movement of the player. Moreover, we discussed unusual bodily orientations and positions with regard to the space or other participants. The TRX was seen as "an alteration mechanism to gravity", "hijacking" the usual bodily positioning, which was considered an interesting feature of the TRX as physical play mediator

Analysis

This case investigated what it means to use embodied sketching very early in the design process, as a way to explore and open up the design space. Before, we described our holistic approach to design, which considers design resources that go beyond the technology, including the socio-spatial configuration of players and other artefacts. Bodystorming is useful to physically sketch ideas that consider different arrangements of these design resources, improving our understanding of the way they intertwine with one another to form design constructs, which we called *embodied sketches*. The activity allows for a rapid arrangement and re-arrangement of resources into different embodied sketches that are tried out, keeping an eye open to how these generate interesting experiential qualities.

This activity resembles traditional brainstorming in the sense that it is done within design groups to come up with ideas on-the-fly, jointly, and includes building on one

another's ideas. However, it does so while also supplying an embodied experience of each idea. Just as in brainstorming, this early form of embodied sketching is done without technology. This method relies on the evocative power of the artefacts, which the participants use to simulate the behaviour of the technology, as well as on the physical core mechanic they invite. This shifts the locus of design attention from the technology to the activity, allowing designers to think about novel ways of using different arrangements of design resources in service of the activity. The workshop does result in the creation of entire games, but on the development of more or less complex 'physical core mechanics' instead.

Example 2: Participatory embodied sketching.

This case example originates in an iteration of a design process in which we were consulting for Movinto Fun, a company that was developing a movement-based game platform, the Oriboo. The device was a ball-shaped interactive toy running on a leash that a participant would hold by its ends. It could sense the players' movements, and it produced sound and visual output (by means of its LED eyes in the front and a small display at the back). The toy was meant to encourage children to explore and expand their bodily movements' repertoire. However, the already implemented games had not met this expectation: firstly, the games encouraged an artefact-focused type of interaction [74], relying heavily on the device's small display. Secondly, the repertoire of movements that the games required was very restricted due to technological limitations [47].

At the point in the design process when this embodied sketching session was organized, we had already made a first attempt at designing a richer repertoire of games. In particular, one of the designed games focused on recognizing movement qualities and had showed potential in earlier play tests [47]. The embodied sketching workshop focused on how this game, as well as one of the original games, could be enriched by using different play patterns and socio-spatial arrangements of players. We also wanted to see how children responded to design constructs that were more or less sustained by other design resources than the technology.

Method

We organized a series of workshops, in which we first let children play freely with the mobile platform, to then play variations of the selected games. The variations made use of the implemented functions in the device, but introduced new socially agreed rules and changed the socio-spatial configuration of the players. Variations included playing one of the original games in a group sharing one Oriboo (instead of having one per child), playing a game blindfolded, or competing between groups. A detailed video analysis was done of the recorded sessions, complemented with in-between interviews where we asked for suggestions.



Fig. 2. Participatory embodied sketching. Multiplayer variation of an original Oriboo game.

Researchers modifying the games on-the-fly

Since the game variations were mostly achieved through socio-spatial and rules modifications, designers could gauge what was working for the players, and what was not, and tweak the games on-the-fly. We could add and remove obstacles in the games to level the difficulty that the game presented to the skills of the participants, or change their use of the technology and the rest of the resources by changing the rules.

For example, in a variant in which the children were blindfolded, we used the device's implemented function of emitting different sounds accordingly to the salient movement quality of the child's movement. All children were using the same function on their device, and the task we posed was to find and make groups with those children who were producing the same sound. The first time this variation was tried, the play session ended too quickly without generating the sense of surmounting a challenge. The reason was that many of the children had chosen the same sound, and since they had started the game close to one another they very quickly ended up in a joint group in the centre of the room. By changing the initial spatial placement of the children, spreading them out in the room, and by explicitly asking some children to produce a specific sound, the activity would last longer and provide a more balanced challenge. To this variation, the children reacted with signs of thrill and excitement.

However, in subsequent play sessions of this variation it turned out to be difficult to establish the right level of challenge. Sometimes, one child would happen to start out too far from others with the same sound, and therefore became clueless as to where to go. We ended up helping with clues to orient them in the "right" direction, either by physically adjusting their walking direction, or with voice cues ("I'm hearing a robot here").

Another example came from a situation where the challenge level was raised after the children had played a particular game of the mobile platform for a while. With most games, the Oriboo had implemented feedback both from its eyes and its display, although children would typically use the

latter as it was easier to decode. So as soon as children seemed to have a game under control, they were instructed to turn the device around, and only use the Oriboo's eyes for feedback and guidance. This seemed to amuse the children. An unexpected effect from this change was that it inspired the children to start playing together, in front of the one another, both holding onto the device, but each one looking at one of its sides.

Children changing the rules

While these workshops were partially structured by the variations introduced by the designers, the workshop would also create social play patterns invented entirely by the children, who appropriated the implemented and suggested games to their needs and wishes. This sometimes involved a different type of holding or sharing of the technology, such as the collaborative transformation of a single-player game into a multiplayer game by finding a different way of holding the device. The children also found variations to the rules suggested by the researchers, such as when they turned a turn-taking game into a collaborative one, and suggested additional rules, like eliminating the player who had missed a move. The children also invented their own games based on basic functions of the technology, like when the children started a racing game with the technology based on the way the "Oriboo's body" moves up along the leash when it's held in a vertical position.

Movements and handling that were not accommodated by the technology

The expected way of holding of the device –from the ends of the stretched leash— was important since it had informed the movement detection algorithm of the game platform. However, during the workshops, it was clear that this would need further revisions, given the many unexpected ways of holding the device we observed. For example, children in a particular school grabbed the device from its body (the central ball), leaving the leash hanging to the sides and bringing it close to their ears. This happened during a game that primarily was based on sound output, in a school where the noise level was quite high. This way of holding the device impeded the children's use of the display as a guide, but influenced the movement recognition system.

Other common types of movements observed included children using the device as a skipping the rope, or using the leash to perform a lasso type of movement (grabbing the ends of the leash) and moving it over their head in a circular movement. All these movements, although incredibly fun for the children, did not fall within what the technology could sense, and therefore reward.

Different usage

Although it is no surprise that different target groups react in different ways to the same technology, and that different contexts in which the activity takes place also influence their reaction, it is worthwhile to understand the intricacies of such differences. In our case, we found that children at different schools found different ways of holding the device, and we were also frequently able to identify why that was the case, like in the above holding of the device close to their ear in a noisy environment. We also saw that certain implemented functions inspired different ways to hold the device in an attempt to 'trick' the device to recognize movements that were not performed. Children from different schools also varied in which core mechanics they preferred [8]. In one school in which the students characterized themselves as very competitive (during post-interviews), competitive games were seen to take most of their attention and enthusiasm. In a school where children had barely had any contact with mobile technology, they were completely engrossed by the interactivity offered by the device, paying most attention to sensorimotor types of games, in particular a game that required repetitive and highly energetic movements. Finally, in a school where children has access to loads of technology devices, and who were very much used to solitary play at home (as reported in questionnaires), they preferred single-player games.

Analysis

This case follows up on the former one in the sense that the focus is still on the interconnection between the design resources that make a particular design sketch. However, this case is less open than the previous one, as we used both implemented functionalities in a working device, and the pre-planned game variations. Still, the workshop format was kept fluent, allowing for innovative variations from both designers and the participants. In this type of embodied sketching, the designers need to act as active observers throughout the workshop, to notice emergent configurations of the social, spatial, technological and artefact resources of the setting.

The decision to leave the designs open for interventions was influenced by the ideas underlying the New Games movement [10] and Wilson's concept of "deputizing the players" [82], both in which the experience of the participants was prioritized over the consistency or stability of the design. Both recognize the entitlement of the player to "uphold, reinterpret, and negotiate the rules as provided" [82]. DeKoven [10] suggests multiple ways to incorporate such on-the-fly modifications, some of which involve the active re-configuration of the game elements by the players. While this is typically easy for players when they feel "at home", playing with their peers [21], it is less likely to happen in the context of an experiment or evaluation. We believe that the fact that the workshop organizers themselves were constantly changing the rules might have been perceived by the children as an invitation to do the same. Finally, the level of technological intervention in this form of embodied sketching is a design choice that can range from functional prototypes to mock-ups, as long as the technology is deployed as one of the many elements that sustain the play activity. This facilitates the capacity of the designer and participants to change the configurations and the rules. In cases where the technology is the only (or the

major) factor responsible for sustaining the game [79], the capacity to influence the activity is lowered.

Example 3: Sensitizing designers

Our third case example was done as a follow-up on the previous workshop around the core mechanic of hanging. Where the former had been useful in creating an understanding of how artefacts of different shape and function would contribute with a range of "working" game ideas, the experience of hanging as such had been little explored. We were yet not able to fully articulate what it was in hanging that was interesting and exhilarating. So this third workshop was developed as a way to delve deeper into that particular experiential concept, and turned to a first-person perspective kind of inquiry.

Method

This workshop resonates with the goal of the embodied storming method by Schleicher et al.: *we ought to first create the experience of physical performance, not to ideate but to enact experiential awareness* [66]. The difference is that we place a higher emphasis on the context of the experience in terms of the physical space, the artefact in use, and the social interrelationships between participants, looking into how these interrelated with the first person experience.

The workshop was done in the form of an AntiGravity (AG) Fundamentals class. AntiGravity Fitness is a pragmatic somaesthetic discipline that comprises different branches or techniques. All of them share a core set of principles and a philosophy, and the use of a physical artefact, the hammock, used in all poses and movements. Many of these involved the participants hanging from the hammock, either in a vertical upright position, or in an inversion with the head towards the floor. Six design researchers were involved. Of those, four are co-authors of this paper, one of which is also a certified AG Fitness instructor who acted as a facilitator and instructed the rest. After the class, we held a brainstorming session with the goal of uncovering interesting experiential qualities and concepts that arose from the experience, which we wanted to relate with elements that sustained the activity, including spatial and social aspects, as well as the hammock itself.



Fig. 3. Sensitizing designers in an AntiGravity class.

An abridge of the experience

The AG class ran as usual AG beginners classes do. Although the class was planned in advance, the instructor changed the poses, the time spent on them, and the feedback and cueing strategy according to her perception of the skills and reactions of the participants. In the following, we illustrate this session with snippets from their impressions collected a few days after the class. *I was so surprised that I was actually able to perform the movements. I watched myself in the mirror and I felt everything was so aesthetically pleasant, my own movements just like the instructor's movements. That felt really empowering*. Another participant recollected the experience as *All excitement and joy*, and elaborated: *It's almost like experiencing flying, even though you know it is not real. It also had that feeling of uncertainty, when you are not sure if you are doing it right or not. You constantly ask and challenge yourself should I do it or should I wait to get the right instruction? It's like always to want more, and pushing forward.*

Debriefing

The analysis of the debriefing of the session resulted in several themes that were related to each other. A salient theme related to the *felt experience* of the activity: AG was empowering in the sense that allowed us to perform movements that we thought we would not be able to. Moreover, it made us reflect on the risk and the safety issues in relation to the practice and ourselves. We also reflected on our abilities and how they relate to the experience of belonging, feeling *in* the activity if we were able to perform an exercise, whereas *out* of it if we could not keep up. This was related to *social aspect* of the activity: although some were self-conscious about their bodies and abilities, we reached out for social support from the rest of the participants and the instructor while performing, wither it was for guidance or validation. When we did not perform an exercise correctly, we felt separated from the rest of the group that could.

Another theme was the *aesthetic component* of the activity. We perceived AG as a deeply aesthetic experience, which we related to the sensed flow of movements, the perception of ourselves and others in synchrony. The soft, mouldable, and yet firm material of the hammock also contributed, and how we could move in it, with it, despite of it, and because of it. It was apparent how the hammock could be shaped, but as much as how it could shape in turn the whole activity and our bodies, triggering bodily reactions, like sweat or a flush in the face, or how it trapped and compressed parts of our bodies, while leaving others free.

Analysis

This case example focuses on the use of a shared lived experience to sensitize designers to salient attributes of their in-the-moment, felt experience, which can then be used in

design. In particular, we harnessed an "unusual" type of bodily activity to elicit creative thinking. In comparison with our other examples, this one is different in that it separates the act of feeling from the actual design activity. The separation was in part a necessity, due to the relative complexity of the movements as such, such as in how the feet must lock around the hammock to be able to safely hang upside down. The second motivation was our experience from the first workshop, which was successful in developing ideas but less successful in creating a good understanding of the felt experience of hanging; We saw a need to separate sensitization from ideation to allow ourselves to focus and access the quite unusual bodily experience of hanging.

The use of the hammocks as a sensitizing aid resonates with previous works in which the concept of *defamiliarization* is used as a tool to trigger creativity. Djajadiningrat et al. [13] describe the method of *interaction relabelling*, where a mechanical artefact is used to evoke possible ways of handling and interacting with a future design in an entirely unforeseen manner –in their paper, they used a toy gun to think about how to interact with an appointment management system. More closely related to our approach is Wilde's method of defamiliarization, in which she encouraged participants to move in different ways than usual [80]. But by focusing on deeply accessing the bodily experience, we take a more pre-ontological stance [49]. For this we use a practice within practical somaesthetics to allow us to sense certain qualities that have potential, and find ways to articulate these. The joint post-processing session was in this sense very important, as it allowed us to abstract from our experiences into aspects that were generalizable as well as shared between us. Finally, these were picked up upon in the design debriefing and subsequent analysis. It is important to highlight the role of the facilitator in this case example. The facilitator helped the rest to articulate, share, and structure their experiences into something that provided a useful context for design.

REFLECTIONS

The three cases illustrate three different ways to use embodied sketching in practice, for different purposes, and with different participants involved: designers/researchers in case 1, and 3, and also end users in case 2. Then, although all three instantiations are to be used early in the design process, they might fit best at different points, given their requirements. For example, while cases 1 and 3 can happen before any other design activity, case 2 require certain design ideas in place, more or less established.

During our work, we have used a bodystorming sessions (case 1) early in design projects, to invite designers to play with design resources and create novel play activity concepts. Once these start to manifest in designed rules over players, artefacts, and technological systems, an embodied sketch emerges, which in turn can inspire other modes of engagement with it. This process is clearly visible in the first case, but emerges also in the participatory setting (case 2), where participants understood and followed, but also tweaked the game structures previously designed by us, and created their own. Maintaining an open and playful approach to subsequent explorations allows us to capture, evaluate and build upon emergent ideas throughout a design project.

The embodied sketching techniques work well together; bodystorming (case 1) and participatory design (case 2) have been used in consecutive workshops in the same design project. Sensitizing workshops (case 3) work well to prepare for bodystorming (case 1), and both should ideally be used at an early stage of a design project. While the bodystorming method succeeds in eliciting design ideas, focusing on physical core mechanics and foregrounding the social aspect of play, it is less sensitive to the felt experience. This was apparent from the discussion notes gathered from the debrief after the workshop, which sometimes would mention experiential qualities but never fully explored them, nor their implications for design. This contrasts with the sensitizing designers workshop, where we were able to access and articulate them in a rich manner.

We argue that such articulation is important to inform designs so that that are in line with, and supportive of the explored felt experience. A related example can be found in [30], in which the group of designers were involved in the practice of Feldenkrais, inspired in which, and for which, they designed.

Five characterizing and distinguishing principles

Our embodied sketching design practice is similar to other embodied ideation methods presented in the background of this paper in that they all use an embodied approach to design, given that they harness the situated physical, and social context for creating and gauging design ideas. However, the goal of embodied sketching, the detailed method, and the resources used, are different.

Five principles characterize and differentiate embodied sketching from other embodied ideation methods: i) an activity-centred embodied approach to ideation, ii) the use of the complete setting as a design resource, iii) the physical and hands-on engagement of designers with a non-scripted activity, iv) the use of movement and play both as method and goal, and v) the facilitation of a sensitizing and design conducive space.

Embodied sketching is activity-centred (i), in that its goal is to design enjoyable social and physical activities that are technologically-supported. By contrast, other embodied approaches to stimulate creativity are artefact-/technology-/service- centred, so that the design activity revolves around designing, prototyping and even testing some*thing* concrete (e.g. Schleicher et al.'s example of designing a new food order system for a cafe). A second important difference is that in embodied sketching, contextual elements are used as design resources, while other embodied methods use

context as a backdrop against which to evaluate design ideas (ii). In addition, although they all are to be used early in the design process, previous methods emphasize early prototype testing (bodystorming, use-case theatre, strong prototyping). In contrast, we focus on initial explorative ideation phases as a way to open up the design space (v) at the experiential level.

For this, we use physical and playful engagement (iv), which we argue to be particularly useful, as it allows designers to engage in a mind-set conducive of exploration and creativity. This is unique to our proposal; others use performative techniques (e.g. acting roles in use-case theatre), or fictional scenarios (e.g. strong prototyping relies on building a high-fi replica of the environment to test mock-ups under different controlled environmental variables, and use-case theatre requires of actors as well as less accurate environment modelling to understand people's possible accommodation and reaction to a particular problem setting and solution). We suggest to use play and playfulness as the primary mode of engagement in all versions of embodied sketching. The advantage of promoting play lies in its transformative and creative power, very similar to those of improvisational theatre but more appropriate for physical and embodied creative activities. But it is also important to realize that we used play to also *design* play and games. This made it easy for our participants to adopt a playful approach to their involvement.

FINAL COMMENTS

In this paper, we have proposed *embodied sketching* to support idea generation when designing for bodily activities in co-located settings where the spatial and physical layout of the environment plays an important role.

While a rigorous comparison between design practices would not be feasible as the goals, method, and outcomes of previous methods and ours are very different, we can reflect upon our results in the light of documented assessments of previous ideation methods. The assessment we have found is Oulasvirta et al.'s, who questioned their success since they did not find that ideas from their embodied methods were more sensitive to the "physical, social or interactional details" compared to those coming from traditional brainstorming methods, and that acting was seen as an "unnecessary factor frustrating participants" [57:7]. We argue that it is harder to motivate and engage with a physical activity during the ideation processes when it appears to be just an aftereffect of the ideation method (e.g. enacting a role), detached from the design goals/outcome. This is remarkably different in embodied sketching, where the activity that participants engage with is meaningful, as it is very close to the final design outcome: you play and move together with your peers to design for play and movement-based social activities (iv).

An important aspect for future work is how to encourage a playful mode of engagement in non-playful design contexts, since playful and physical engagement might pose a challenge for certain participants and in certain design domains. We didn't experience such difficulties in the design cases presented in this paper, but they all belong to a very particular design domain (physical play). Strategies for design projects outside this domain remain as very relevant future work, and we are currently working towards developing them.

Another point to consider for future work is to study the impact that our design practices have on both the participants and the outcome of ideation process, which we can only do by exposing our practices to multiple design contexts.

Finally, we want to conclude by highlighting a particular feature of our embodied sketches: Physical activity is the defining and critical aspect of our embodied sketching practice. It is the in-the-moment activity that shapes both the opportunities for uncovering and scrutinizing the felt experience, and the fluid and flexible space for creativity, innovation, and change. This means that the sketches obtained are not persistent but *ephemeral*, shaped in the moment and then gone. A critical issue for embodied interaction concerns how to enable participants to remember, document, and articulate these ephemeral sketches. In our case examples, this was primarily done through extensive video documentation. All our sessions were filmed, often from multiple directions with multiple cameras. Participants were also asked to talk about their experience right after the embodied sketching session, so as to capture their thoughts and experiences as rapidly and clearly as possible. It remains as future work to develop ways of packaging the observed design features of the ephemeral sketches in a way that becomes readily available to the designers.

ACKNOWLEDGMENTS

Many people have made this paper possible. First, the colleagues who participated in our workshops: Jon Back, Mareike Glöss, Floyd Mueller, Robert Cercós, Jayden Garner, William Goddard, and Harrison Smith, and the children from Ullens (Kathmandu), Rudrayanee (Kathmandu), Sophia (Stockholm), and Rico Cejudo (Seville) schools. Special thanks to those who facilitated our workshops: Medin Lamichhane (Ullens), Babu Ram Thapa (Rudrayanee), Louise Öhnfeldt (Sophia), Lola Segura (Rico Cejudo), and Yoram Chisik. Also to Jessica Söderlund, who lend us her fitness studio Yogaire for our sensitizing workshop. Thanks to our helpers Carlos Duarte, Adrián Onco, Luis Maqueda, Mathias Jonsson, and to Jin Moen, who shipped us worldwide with a bunch of Oriboos. Thanks to the CHI reviewers, who provided very helpful comments, which have helped polish our contributions, and to Donny McMillan who proofread this text. Part of the research was made possible by a grant from the Swedish Governmental Agency for Innovation Systems (VINNOVA) to the Mobile Life Vinn Excellence Center.

REFERENCES

1. Jeffrey Bardzell. 2009. Interaction Criticism and Aesthetics. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '09), 2357–2366. http://doi.org/10.1145/1518701.1519063
2. Steve Benford, Holger Schnädelbach, Boriana Koleva, et al. 2005. Expected, Sensed, and Desired: A Framework for Designing Sensing-based Interaction. *ACM Trans. Comput.-Hum. Interact.* 12, 1: 3–30. http://doi.org/10.1145/1057237.1057239
3. Julian Bleecker. 2009. Design Fiction: A Short Essay on Design, Science, Fact and Fiction. *Near Future Laboratory*. Retrieved September 25, 2015 from http://blog.nearfuturelaboratory.com/2009/03/17/design-fiction-a-short-essay-on-design-science-fact-and-fiction/
4. Mark Blythe. 2014. The Hitchhiker's Guide to Ubicomp: Using Techniques from Literary and Critical Theory to Reframe Scientific Agendas. *Personal Ubiquitous Comput.* 18, 4: 795–808. http://doi.org/10.1007/s00779-013-0679-6
5. Jacob Buur and Rosa Torguet. 2013. Ethnographic findings in the organizational theatre. *Ethnographic Praxis in Industry* 1: 143–160. http://doi.org/10.1111/j.1559-8918.2013.00013.x
6. Bill Buxton. 2007. *Sketching User Experiences: Getting the Design Right and the Right Design*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
7. Diane Carr, Gareth Schott, Andrew Burn, and David Buckingham. 2004. Doing game studies: A multi-method approach to the study of textuality, interactivity and narrative space. *Media International Australia* 110: 19–30.
8. Yoram I. Chisik, Alissa N. Antle, Brian Birtles, Elena Márquez Segura, and Cristina Sylla. 2014. The Kathmandu Kids Entertainment Workshops. In *Entertaining the Whole World*, Adrian David Cheok, Anton Nijholt and Teresa Romão (eds.). Springer London, 5–21.
9. Yvonne A. W. De Kort and Wijnand A. Ijsselsteijn. 2008. People, Places, and Play: Player Experience in a Socio-spatial Context. *Computers in Entertainment.* 6, 2: 18:1–11. http://doi.org/10.1145/1371216.1371221
10. Bernard De Koven. Coliberation. *DeepFUN*. Retrieved September 25, 2015 from http://www.deepfun.com/coliberation/
11. Sebastian Deterding. 2009. The Game Frame: Systemizing a Goffmanian Approach to Video Game Theory. *Digital Games Research Association.*
12. Sebastian Deterding. 2011. Meaningful Play. Getting »Gamification« Right. In *Google Tech Talk.* Retrieved December 18, 2015 from http://www.slideshare.net/dings/meaningful-play-getting-gamification-right
13. J. P. Djajadiningrat, W. W. Gaver, and J. W. Frens. 2000. Interaction Relabelling and Extreme Characters: Methods for Exploring Aesthetic Interactions. In *Proceedings of the 3rd Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (DIS 00), 66–71. http://doi.org/10.1145/347642.347664
14. Tom Djajadiningrat, Stephan Wensveen, Joep Frens, and Kees Overbeeke. 2004. Tangible Products: Redressing the Balance Between Appearance and Action. *Personal Ubiquitous Comput.* 8, 5: 294–309. http://doi.org/10.1007/s00779-004-0293-8.
15. Andrei Dobra. Great Games Are Needed to Make PlayStation Move Popular, Sony Admits. In *Softpedia*. Retrieved June 7, 2015 from http://news.softpedia.com/news/Great-Games-Are-Needed-to-Make-PlayStation-Move-Popular-Sony-Admits-257838.shtml
16. Paul Dourish. 2001. *Where the Action Is: The Foundations of Embodied Interaction*. The MIT Press, Cambridge, Mass.
17. Nicolas Ducheneaut, Nicholas Yee, Eric Nickell, and Robert J. Moore. 2006. "Alone Together?": Exploring the Social Dynamics of Massively Multiplayer Online Games. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '06), 407–416. http://doi.org/10.1145/1124772.1124834.
18. Daniel Fallman and Erik Stolterman. 2010. Establishing Criteria of Rigor and Relevance in Interaction Design Research. In *Proceedings of the 2010 International Conference on The Interaction Design*, (Create '10), Oli Mival, John Bonner, Michael Smyth, and Shaleph O'Neill (eds.). British Computer Society, Swinton, UK, 58-63.
19. Greta G. Fein. 1981. Pretend Play in Childhood: An Integrative Review. *Child Development* 52, 4: 1095–1118. http://doi.org/10.2307/1129497
20. James Fogarty, Jodi Forlizzi, and Scott E. Hudson. 2001. Aesthetic Information Collages: Generating Decorative Displays That Contain Information. in *Proceedings of the 14th Annual ACM Symposium on User Interface Software and Technology* (UIST '01), 141–150. http://doi.org/10.1145/502348.502369
21. Tracy Fullerton. 2008. *Game Design Workshop: A Playcentric Approach to Creating Innovative Games*. Morgan Kaufmann, Amsterdam; Boston.
22. Brian Gajadhar, Yvonne de Kort, and Wijnand IJsselsteijn. 2008. Influence of Social Setting on Player Experience of Digital Games. In *CHI'08 Extended Abstracts on Human Factors in Computing Systems* (CHI EA '08), 3099–3104. http://doi.org/10.1145/1358628.1358814
23. Bill Gaver. 2002. Designing for Homo Ludens. *I3 Magazine 12*. Retrieved January 24, 2016 from https://www.gold.ac.uk/media/migrated/media/goldsmiths/departments/researchcentres/interactionresearchstudio/pdf/27gaver.ludens.02.pdf
24. Elizabeth Gerber. 2007. Improvisation Principles and Techniques for Design. In *Proceedings of the SIGCHI*

Conference on Human Factors in Computing Systems (CHI '07), 1069–1072. http://doi.org/10.1145/1240624.1240786
25. Elizabeth Gerber. 2009. Using Improvisation to Enhance the Effectiveness of Brainstorming. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI ' 09), 97–104. http://doi.org/10.1145/1518701.1518718
26. Erving Goffman. 1961. Fun in Games. *Encounters: Two studies in the sociology of interaction*. Bobbs-Merrill, Oxford, England, 15 – 81.
27. Dave Gray, Sunni Brown, and James Macanufo. 2010. *Gamestorming: A Playbook for Innovators, Rulebreakers, and Changemakers*. O'Reilly Media Inc., Sebastopol, CA, USA.
28. Marc Hassenzahl. 2008. The Interplay of Beauty, Goodness, and Usability in Interactive Products. *Hum.-Comput. Interact.* 19, 4: 319–349. http://doi.org/10.1207/s15327051hci1904_2
29. Kristina Höök. 2010. Transferring Qualities from Horseback Riding to Design. *Proceedings of the 6th Nordic Conference on Human-Computer Interaction: Extending Boundaries* (NordiCHI '10), ACM, 226–235. http://doi.org/10.1145/1868914.1868943
30. Kristina Höök, Anna Ståhl, Martin Jonsson, Johanna Mercurio, Anna Karlsson, and Eva-Carin Banka Johnson. 2015. Somaesthetic Design. *Interactions* 22, 4: 26–33. http://doi.org/10.1145/2770888
31. Johan Huizinga. 1971. *Homo Ludens: A Study of the Play-Element in Culture*. Beacon Press, Boston.
32. Caroline Hummels, Kees C. Overbeeke, and Sietske Klooster. 2007. Move to Get Moved: A Search for Methods, Tools and Knowledge to Design for Expressive and Rich Movement-based Interaction. *Personal Ubiquitous Comput.* 11, 8: 677–690. http://doi.org/10.1007/s00779-006-0135-y
33. Giulio Iacucci, Carlo Iacucci, and Kari Kuutti. 2002. Imagining and Experiencing in Design, the Role of Performances. *Proceedings of the Second Nordic Conference on Human-computer Interaction* (NordiCHI '02), 167–176. http://doi.org/10.1145/572020.572040
34. Giulio Iacucci and Kari Kuutti. 2002. Everyday Life as a Stage in Creating and Performing Scenarios for Wireless Devices. *Personal Ubiquitous Comput.* 6, 4: 299–306. http://doi.org/10.1007/s007790200031
35. Katherine Isbister and Kaho Abe. 2015. Costumes As Game Controllers: An Exploration of Wearables to Suit Social Play. *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction* (TEI '15), 691–696. http://doi.org/10.1145/2677199.2688813
36. Katherine Isbister, Rahul Rao, Ulf Schwekendiek, Elizabeth Hayward, and Jessamyn Lidasan. 2011. Is More Movement Better?: A Controlled Comparison of Movement-based Games. *Proceedings of the 6th International Conference on Foundations of Digital Games* (FDG ' 11), 331–333. http://doi.org/10.1145/2159365.2159429
37. Esther Jakobs, Antony S. R. Manstead, and Agneta H. Fischer. 1996. Social context and the experience of emotion. *Journal of Nonverbal Behavior* 20, 2: 123–142. http://doi.org/10.1007/BF02253073
38. Wonjun Lee, Youn-kyung Lim, and Richard Shusterman. 2014. Practicing Somaesthetics: Exploring Its Impact on Interactive Product Design Ideation. *Proceedings of the 2014 Conference on Designing Interactive Systems* (DIS '14), 1055–1064. http://doi.org/10.1145/2598510.2598561
39. Lian Loke and George Poonkhin Khut. 2011. Surging Verticality: An Experience of Balance. *Proceedings of the Fifth International Conference on Tangible, Embedded, and Embodied Interaction* (TEI ' 11), 237–240. http://doi.org/10.1145/1935701.1935747
40. Lian Loke, Astrid T. Larssen, Toni Robertson, and Jenny Edwards. 2007. Understanding Movement for Interaction Design: Frameworks and Approaches. *Personal Ubiquitous Comput.* 11, 8: 691–701. http://doi.org/10.1007/s00779-006-0132-1
41. Lian Loke and Toni Robertson. 2011. The Lived Body in Design: Mapping the Terrain. In *Proceedings of the 23rd Australian Computer-Human Interaction Conference* (OzCHI '11), 181–184. http://doi.org/10.1145/2071536.2071565
42. Lian Loke and Toni Robertson. 2013. Moving and Making Strange: An Embodied Approach to Movement-based Interaction Design. *ACM Trans. Comput.-Hum. Interact.* 20, 1: 7:1–7:25. http://doi.org/10.1145/2442106.2442113
43. Catriona Macaulay, Giulio Jacucci, Shaleph O'Neill, Tomi Kankaineen, and Morna Simpson. 2006. Editorial: The Emerging Roles of Performance Within HCI and Interaction Design. *Interact. Comput.* 18, 5: 942–955. http://doi.org/10.1016/j.intcom.2006.07.001
44. Carsten Magerkurth, Timo Engelke, and Maral Memisoglu. 2004. Augmenting the Virtual Domain with Physical and Social Elements: Towards a Paradigm Shift in Computer Entertainment Technology. *Comput. Entertain.* 2, 4: 12–12. http://doi.org/10.1145/1037851.1037870
45. Antony Stephen Reid Manstead. 2005. The social dimension of emotion. *The Psychologist* 18, 8: 484–487.
46. Elena Márquez Segura and Katherine Isbister. 2015. Enabling Co-Located Physical Social Play: A Framework for Design and Evaluation. In *Game User Experience Evaluation*, Regina Bernhaupt (ed.). Springer International Publishing, 209–238.
47. Elena Márquez Segura, Annika Waern, Jin Moen, and Carolina Johansson. 2013. The Design Space of Body Games: Technological, Physical, and Social Design. In In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '13), 3365–3374. http://doi.org/10.1145/2470654.2466461

48. Helena M. Mentis, Kristina Höök, Florian Mueller, Katherine Isbister, George Poonkhin Khut, and Toni Robertson. 2014. Designing for the Experiential Body. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems* (CHI EA '14), 1069–1074. http://doi.org/10.1145/2559206.2579402
49. Maurice Merleau-Ponty. 2002. *Phenomenology of Perception*. Psychology Press.
50. Angela Moscaritolo. Microsoft: Xbox One Sales Double After Dropping Kinect. *PCMAG*. Retrieved June 7, 2015 from http://www.pcmag.com/article2/0,2817,2461023,00.asp
51. Florian Mueller, Stefan Agamanolis, and Rosalind Picard. 2003. Exertion Interfaces: Sports over a Distance for Social Bonding and Fun. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '03), 561–568. http://doi.org/10.1145/642611.642709
52. Florian Mueller, Cagdas Toprak, Eberhard Graether, Wouter Walmink, Bert Bongers, and Elise van den Hoven. 2012. Hanging off a Bar. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems* (CHI EA '12), 1055–1058. http://doi.org/10.1145/2212776.2212384
53. A. F. Newell, M. E. Morgan, P. Gregor, and A. Carmichael. 2006. Theatre As an Intermediary Between Users and CHI Designers. In *CHI '06 Extended Abstracts on Human Factors in Computing Systems* (CHI EA ' 06), 111–116. http://doi.org/10.1145/1125451.1125479
54. Lene Nielsen. 2002. From User to Character: An Investigation into User-descriptions in Scenarios. In *Proceedings of the 4th Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (DIS '02), 99–104. http://doi.org/10.1145/778712.778729
55. Kenton O'hara, Richard Harper, Helena Mentis, Abigail Sellen, and Alex Taylor. 2013. On the Naturalness of Touchless: Putting the "Interaction" Back into NUI. *ACM Trans. Comput.-Hum. Interact.* 20, 1: 5:1–5:25. http://doi.org/10.1145/2442106.2442111
56. Alex F. Osborn and Lee Hastings Bristol. 1979. *Applied Imagination: Principles and Procedures of Creative Thinking*. Charles Scribner's Sons, New York.
57. Antti Oulasvirta, Esko Kurvinen, and Tomi Kankainen. 2003. Understanding Contexts by Being There: Case Studies in Bodystorming. *Personal Ubiquitous Comput.* 7, 2: 125–134. http://doi.org/10.1007/s00779-003-0238-7
58. Marianne Graves Petersen, Ole Sejer Iversen, Peter Gall Krogh, and Martin Ludvigsen. 2004. Aesthetic Interaction: A Pragmatist's Aesthetics of Interactive Systems. In *Proceedings of the 5th Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (DIS '04), 269–276. http://doi.org/10.1145/1013115.1013153
59. Cindy Poremba. 2007. Critical Potential on the Brink of the Magic Circle. Retrieved January 24, 2016 from http://www.digra.org/wp-content/uploads/digital-library/07311.42117.pdf
60. Jenny Preece, Yvonne Rogers, and Helen Sharp. 2015. *Interaction Design: Beyond Human-Computer Interaction*. John Wiley & Sons.
61. Niklas Ravaja, Timo Saari, Marko Turpeinen, Jari Laarni, Mikko Salminen, and Matias Kivikangas. 2006. Spatial Presence and Emotions During Video Game Playing: Does It Matter with Whom You Play? *Presence: Teleoper. Virtual Environ.* 15, 4: 381–392. http://doi.org/10.1162/pres.15.4.381
62. Katie Salen and Eric Zimmerman. 2003. *Rules of Play: Game Design Fundamentals*. The MIT Press, Cambridge, Mass.
63. Thecla Schiphorst. 2007. Really, Really Small: The Palpability of the Invisible. In *Proceedings of the 6th ACM SIGCHI Conference on Creativity & Cognition* (C&C '07), 7–16. http://doi.org/10.1145/1254960.1254962
64. Thecla Schiphorst. 2009. Soft(N): Toward a Somaesthetics of Touch. In *CHI '09 Extended Abstracts on Human Factors in Computing Systems* (CHI EA '09), 2427–2438. http://doi.org/10.1145/1520340.1520345
65. Thecla Schiphorst. 2011. Self-evidence: applying somatic connoisseurship to experience design. *CHI '11 Extended Abstracts on Human Factors in Computing Systems* (CHI EA '11), 145–160. http://doi.org/10.1145/1979742.1979640
66. Dennis Schleicher, Peter Jones, and Oksana Kachur. 2010. Bodystorming As Embodied Designing. *Interactions* 17, 6: 47–51. http://doi.org/10.1145/1865245.1865256
67. Richard Shusterman. 2008. *Body Consciousness: A Philosophy of Mindfulness and Somaesthetics*. Cambridge University Press, Cambridge ; New York.
68. Richard Shusterman. 2013. Somaesthetics. *The Encyclopedia of Human-Computer Interaction, 2nd Ed*. Retrieved September 25, 2015 from https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/somaesthetics
69. Bart Simon. 2009. *Wii are Out of Control: Bodies, Game Screens and the Production of Gestural Excess*. Social Science Research Network, Rochester, NY. Retrieved June 7, 2015 from http://papers.ssrn.com/abstract=1354043
70. Dag Svanaes. 2013. Interaction Design for and with the Lived Body: Some Implications of Merleau-ponty's Phenomenology. *ACM Trans. Comput.-Hum. Interact.* 20, 1: 8:1–8:30. http://doi.org/10.1145/2442106.2442114

71. Joe Svetlik. 2012. Sony admits selection of PS Move games "not so great." *CNET*. Retrieved June 7, 2015 from http://www.cnet.com/news/sony-admits-selection-of-ps-move-games-not-so-great/
72. Christine Szentgyorgyi, Michael Terry, and Edward Lank. 2008. Renegade Gaming: Practices Surrounding Social Use of the Nintendo DS Handheld Gaming System. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '08), 1463–1472. http://doi.org/10.1145/1357054.1357283
73. Joshua Tanenbaum and Karen Tanenbaum. 2015. Envisioning the Future of Wearable Play: Conceptual Models for Props and Costumes as Game Controllers. In Proceedings of the 10th International Conference on the Foundations of Digital Games (FDG '15).
74. Jakob Tholander and Carolina Johansson. 2010. Design Qualities for Whole Body Interaction: Learning from Golf, Skateboarding and BodyBugging. In *Proceedings of the 6th Nordic Conference on Human-Computer Interaction: Extending Boundaries* (NordiCHI '10), 493–502. http://doi.org/10.1145/1868914.1868970
75. TRX. 2016. Retrieved January 24, 2016 from https://www.trxtraining.com/
76. Sherry Turkle. 2012. *Alone Together: Why We Expect More from Technology and Less from Each Other*. Basic Books, New York, NY.
77. John Vines, Tess Denman-Cleaver, Paul Dunphy, Peter Wright, and Patrick Olivier. 2014. Experience Design Theatre: Exploring the Role of Live Theatre in Scaffolding Design Dialogues. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '14), 683–692. http://doi.org/10.1145/2556288.2556960
78. Amy Voida and Saul Greenberg. 2009. Wii All Play: The Console Game As a Computational Meeting Place. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '09), 1559–1568. http://doi.org/10.1145/1518701.1518940
79. Annika Waern. 2009. Information Technology in Pervasive Games. In *Pervasive Games*, Markus Montola, Jaakko Stenros and Annika Waern (eds.). Morgan Kaufmann, Boston, 163–174.
80. Danielle Wilde. 2008. The Hipdiskettes: Learning (Through) Wearables. *Proceedings of the 20th Australasian Conference on Computer-Human Interaction: Designing for Habitus and Habitat* (OzCHI '08), 259–262. http://doi.org/10.1145/1517744.1517771
81. Danielle Wilde, Thecla Schiphorst, and Sietske Klooster. 2011. Move to Design/Design to Move: A Conversation About Designing for the Body. *Interactions* 18, 4: 22–27. http://doi.org/10.1145/1978822.1978828
82. Douglas Wilson. 2011. Brutally Unfair Tactics Totally OK Now: On Self-Effacing Games and Unachievements. *Game Studies* 11, 1. Retrieved September 25, 2015 from http://gamestudies.org/1101/articles/wilson